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Application management and effectiveness analysis of intelligent logistics robots in hospital drug and specimen delivery scenarios

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To address the limitations of traditional manual logistics in hospitals, this study evaluated an intelligent logistics robot system for drug and specimen delivery. We conducted a six-month single-center parallel-controlled study in a tertiary hospital. The system consisted of 10 autonomous mobile robots using a hierarchical algorithm framework: Dijkstra's algorithm for global path planning, A* algorithm with Dynamic Window Approach for local navigation, and M/M/c queuing model for multi-robot coordination. Compared with manual delivery, robot delivery reduced delivery time by 32% to 36% ($p < 0.001$). The robot group achieved 100% verification accuracy and item integrity rate, with significantly lower omission rate ($p = 0.011$). The 10-robot fleet completed 7.3 times more delivery trips than 19 manual workers over six months. Both nurse and patient satisfaction were significantly higher in the robot group ($p < 0.05$). Over a 10-year lifecycle, the robot system saved 6.928 million RMB, with sensitivity analysis confirming economic robustness. The intelligent logistics robot system demonstrated significant advantages in delivery efficiency, quality, scalability, user satisfaction, and long-term cost-effectiveness.

Keywords Logistics robot, Drug delivery, Specimen transport, Smart hospital construction

The global transformation toward smart hospitals is reshaping healthcare service models. New technologies such as artificial intelligence (AI), the Internet of Things (IoT), and big data are driving this change[1,2]. The intra-hospital logistics system serves as the "lifeline" for efficient and safe hospital operations. It handles the flow of critical materials including drugs, specimens, consumables, and sterile items. Its performance directly affects medical quality, operational efficiency, and patient safety[3].

However, traditional manual logistics faces multiple challenges. First, rising labor costs and declining interest in logistics jobs have created recruitment difficulties and high turnover rates[4,5]. Second, manual operations carry inevitable error risks, such as wrong drug delivery and specimen confusion[4]. Third, capacity bottlenecks during peak hours and long elevator wait times seriously affect delivery timeliness[6]. The COVID-19 pandemic further highlighted the risk of cross-infection from personnel movement and contact[7,8].

Various automation solutions have emerged to address these challenges. Studies show that centralized automated dispensing systems can significantly reduce dispensing error rates. For example, a French teaching hospital reduced its dispensing error rate from 2.9% to 1.7% after introducing such a system[9]. In Chinese tertiary hospitals, the integration of intelligent dispensing robots with logistics robots reduced drug waste by about 59.08%[10]. The application of robots in automated labeling, sorting, and distribution of infusion bags further reduces infection risks associated with manual operations and workloads[11]. From a macro perspective, intelligent logistics systems enhance the visibility and traceability of pharmaceutical supply chains by achieving dual cost benefits through reduced labor requirements, decreased consumption of protective materials, and improved warehouse utilization rates[12,13]. During the pandemic, contactless delivery by robots proved effective in reducing exposure risk for medical staff[14].

Traditional hardware solutions for hospital logistics, such as rail-guided vehicle systems and pneumatic tube systems, have shown potential for improving efficiency[15–17]. However, these solutions share common drawbacks:

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high initial construction costs, need for large-scale infrastructure modifications, poor path flexibility, and limited load capacity. Hospitals urgently need a logistics technology with low infrastructure dependency, high flexibility, and strong transport capacity.

Intelligent logistics robots have emerged to meet this need. They integrate autonomous navigation, multi-sensor fusion, IoT, and cloud computing. Their flexible track-free navigation allows deployment in existing environments without large-scale modifications. Theoretically, such robots can significantly reduce reliance on human labor, enhance delivery accuracy and traceability, effectively minimize interpersonal contact, thereby demonstrating notable advantages in improving efficiency, ensuring safety, and strengthening epidemic prevention and control measures[2,18]. From an operations research perspective, intra-hospital logistics can be modeled as a dynamic Vehicle Routing Problem (VRP) with time windows, capacity constraints, and random demand parameters[19,20]. The core challenge is assigning robot fleets to delivery tasks while minimizing total travel time and ensuring on-time completion[21]. The technologies required for multi-robot collaborative execution of parallel tasks are closely related to multi-agent task allocation problems, which constitute fundamental research topics in distributed artificial intelligence and robotics fields[22,23]. The decentralized coordination protocol adopted in this system embodies core principles of market-based task allocation and consensus algorithms in multi-agent systems by balancing workloads and preventing resource conflicts[24]. Additionally, interactions between robots and shared resources such as elevators, charging stations, and narrow corridors generate queuing phenomena that can be analyzed using queuing theory[25,26]. Delivery request arrival times, service durations at pick-up/delivery points, and elevator availability can all be modeled as queuing networks. By applying queuing theory, we can predict system bottlenecks and optimize system parameters (e.g., fleet size, charging schedules) to ensure stable operation during demand fluctuations[27]. These theoretical frameworks collectively form the foundation for designing and evaluating robotic logistics systems.

Despite the theoretical advantages of intelligent logistics robots, their practical application in healthcare settings remains in an exploratory stage. Current research predominantly focuses on single-task scenarios such as sterile instrument transportation or navigation tasks[28,29], or while involving multitasking, relies on simulation data or preliminary deployment[30,31]. Evaluation metrics often concentrate on time efficiency, lacking comprehensive consideration of quality, user satisfaction, and long-term cost-effectiveness (see Table 1). To address this gap, this study utilizes smart hospital construction as a practical platform to design and validate a multitasking, closed-loop intelligent robotic logistics system for pharmaceutical and specimen delivery. A six-month empirical study was conducted in real hospital environments to systematically evaluate its performance relative to traditional manual delivery across dimensions including efficiency, quality, capacity, user satisfaction, and long-term cost-effectiveness. Specifically, this study aims to address the following questions: Does the logistics robot system outperform manual delivery in terms of efficiency and quality? How does it perform in task processing capabilities and scalability? Can it enhance satisfaction levels among nurses and patients? And is it more cost-effective in the long term? We hypothesize that the robotic system will demonstrate the following advantages: (1) Significant reduction in delivery time; (2) Improved verification accuracy and completeness with reduced error rates; (3) Enhanced task processing capabilities and scalability; (4) Increased user satisfaction; (5) Cost savings over a 10-year lifecycle. This research will provide a scientifically quantifiable basis for intelligent logistics decision-making in large hospitals.

Methods

Study design

This study used a single-center, parallel-controlled observational design. We compared manual delivery with robot delivery over six months from June 1 to November 30, 2025. The study evaluated performance from five dimensions: efficiency, quality, capacity, user satisfaction, and cost-effectiveness.

To ensure comparability between the two groups, we implemented several control measures. (1) Temporal Alignment: Both datasets were collected from the same time period, excluding the influence of seasonal variations and hospital demographic fluctuations. (2) Route Standardization: All delivery routes covered identical origin-

Study	System type	Task mode	Deployment	Evaluation metrics	Scale	Key limitations
Fragapane et al.[32]	AMR	Multi-task	Real-world (Case study)	Qualitative (Benefits, applicability)	2 Hospitals	Preliminary study, limited generalizability, no quantitative metrics
Fragapane et al.[28]	AMR, AGV, Manual	Single-task	Real-world (Multiple case study)	Flexibility, Productivity, Quality/Service, Costs	3 Hospitals	Focus on sterile instruments only, European hospitals context
Cheng et al.[30]	AMR	Multi-task	Simulation	Cost (Fixed, penalty, transportation)	54 Simulated Instances (20, 50, 100 requests)	Assumes normal distribution, all requests known beforehand
Rondoni et al.[29]	AMR	Single-task	Real-world (Simulated hospital)	Completion Time, Path Length, Distance Error, Orientation Error, Success Rate, Min Distance, Time Cruise Speed	2 Robots (HOSBOT, TIAGo)	Small scale, controlled environment, tested only two robots
Valner et al.[31]	Heterogeneous Fleet (AMR)	Multi-task	Real-world (Field test)	Qualitative (Feasibility, lessons learned)	1 Robot (TIAGo) + custom doors	Single robot deployment, limited task scope, reliance on custom infrastructure

Table 1. Comparison of hospital logistics robot studies. AMR, autonomous mobile robot; AGV, automated guided vehicle.

destination pairs (from pharmacy to ward, from ward to laboratory), with consistent distance and path complexity. (3) Workload Balance: Only routine scheduled delivery tasks within standard working hours (8:00–17:30) were included, excluding emergency deliveries, off-duty tasks, and cases with incomplete data records. Preliminary review indicated that excluded cases accounted for less than 5% of total tasks, demonstrating representativeness of routine operational scenarios. (4) Environmental Control: The robotic system is equipped with IoT elevator control modules, enabling priority access equivalent to manual operators. The human team members averaged over 6 months of hospital logistics experience, reflecting stable and well-trained workforce capabilities.

Logistics robot system design and functions

Logistics robot system architecture

The hospital's logistics robot system is an intelligent solution with deep integration of software and hardware. Its architecture consists of three parts: the robot unit, the dispatch software platform, and environmental interaction facilities. (See Fig. 1) The specific design is as follows:

(1) Robot Unit:

a. Intelligent Navigation: It integrates laser SLAM and multi-sensor technology to achieve autonomous path planning, dynamic obstacle avoidance, and global optimal path decision-making.
b. Modular Cargo Compartment: It provides 6 types of standardized compartments, supporting flexible configuration according to department needs. The fully enclosed design ensures transport safety.
c. Identity Verification: It automatically identifies the task initiator and recipient information through RFID technology.
d. Status Monitoring: It transmits location, task progress, and device status in real-time, with immediate alerts for any abnormalities.

(2) Software System:

a. Dispatch Control Platform: It builds a 3D visualized operational scene based on digital twin technology, enabling dynamic task allocation and cluster scheduling.
b. Data Interoperability: It seamlessly integrates with the hospital's HIS, LIS, and SPD systems, deeply mining key indicators like logistics timeliness and resource utilization to generate operational optimization reports.

(3) Environmental Interaction Facilities:

a. IoT Collaboration: It interacts with elevators and automatic access control systems via IoT protocols to ensure continuity of cross-area transport.
b. Autonomous Power Management: It is equipped with smart charging piles that support automatic charging during idle periods.

Core functions of the logistics robot

(1) Intelligent Navigation and Positioning

b. Autonomous Power Management: It is equipped with smart charging piles that support automatic charging during idle periods.

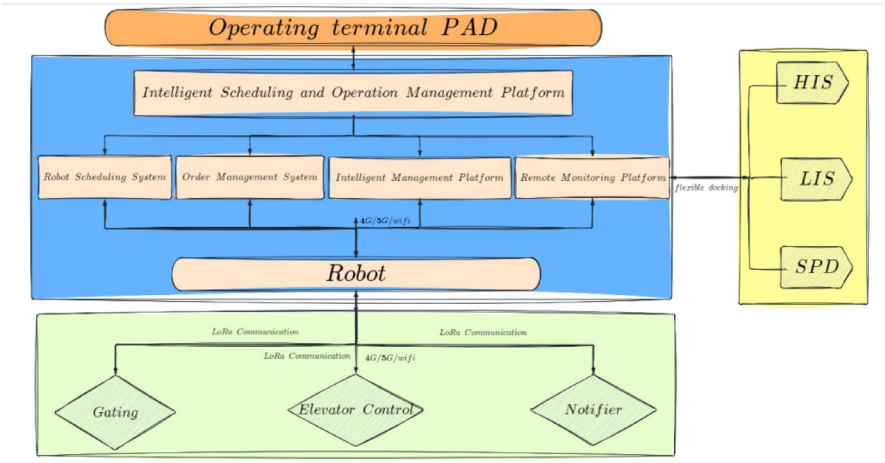


Fig. 1. System architecture.

Based on SLAM technology, the robot perceives and recognizes its surrounding environment. It uses preset algorithms combined with real-time environmental information for path planning and navigation. This includes calculating one or more safe and efficient routes based on the destination, traffic rules, and obstacle locations. It achieves centimeter-level positioning accuracy and dynamic path planning, ensuring autonomous obstacle avoidance and transport safety in complex environments.

(2) Human-Robot Interaction Functions

One-click call: Medical staff or workers can use a PAD or console to make a one-click call to request the robot to deliver items, achieving on-demand responsive material dispatch.

Task assignment: In the application or console, users can input information about a delivery task. After receiving the task, the robot automatically plans its route and starts delivery. Multiple interactive consoles are provided to make it convenient for medical staff to assign tasks at any time.

Identity verification: To prevent staff from placing the wrong items during delivery or taking the wrong items upon receipt, the robot uses an authenticated pickup and delivery design. Medical staff must swipe a card or use a fingerprint for identity verification. Only after verification can they place or retrieve items, ensuring controlled access for pickup and delivery operations.

Multimodal reminders: During delivery, the robot can use its built-in voice system for announcements, notifying recipients about item information or prompting people to make way. In complex crowd environments, it prompts people to avoid it to ensure its safety. When the robot arrives with materials, a notifier at the station will sound to alert medical staff to pick up the items, ensuring the timeliness of material transport.

(3) Environmental Collaboration Capabilities

Automatic door control: By installing an access controller, the robot communicates via the LoRa protocol to achieve automatic door opening and closing, enabling contactless autonomous passage.

Cross-floor delivery: The robot's built-in elevator control system allows it to autonomously control elevators when needed for delivery between different floors.

Destination recognition: The robot can automatically identify the destination by reading the RFID tag on the material transfer box, ensuring precise delivery.

(4) Backend Monitoring and Management

Real-time monitoring: Administrators can use the backend system to monitor the robot's operational status, task progress, battery level, and other key information in real-time. They can also monitor for abnormal states, allowing for prompt and quick handling, and can schedule regular maintenance and repairs. This enables 24-hour uninterrupted maintenance, ensuring the robot's safe operation at all times.

Video traceability: Cameras are installed at pickup and drop-off stations. When medical staff pick up or drop off items, the cameras record a video. If a delivery error occurs, the video records can be reviewed at any time to trace and track the error.

Logistics tracking: The status and progress of the robot's transport tasks can be monitored and tracked in real-time. It can also estimate the material arrival time, reducing anxiety among medical staff about delivery timeliness.

Intelligent power management: When the robot's battery level falls below a certain threshold, it automatically returns to a charging pile to recharge. This prevents delivery work from being affected by a depleted battery and ensures 24-hour continuous operation.

Algorithm framework

The robot dispatch system uses a hierarchical algorithm framework to ensure efficient task execution, collision-free navigation, and optimal resource utilization.

(1) Task prioritization: tasks are sorted by clinical urgency. Priority 1 (urgent): emergency drugs and specimens, immediate dispatch. Priority 2 (routine): regular drugs and specimens, FIFO queue processing. Priority 3 (scheduled): timed deliveries, time-window based scheduling.
(2) Path planning uses a hybrid approach: Global path planning is based on Dijkstra's algorithm to calculate the shortest feasible path; Local navigation uses A* algorithm with dynamic obstacle avoidance; Path optimization combines improved nearest neighbor heuristic with 2-opt local search strategy.
(3) Collision avoidance uses Dynamic Window Approach (DWA) with multi-sensor fusion, maintaining a 0.5 m safety buffer and recalculating velocity commands at 10 Hz frequency. Emergency braking is triggered when obstacles are detected within 0.3 m; Static obstacle avoidance preset restricted zones; Dynamic obstacle avoidance based on LiDAR detection combined with A* path re-planning.
(4) Multi-robot coordination uses nearest-available assignment strategy. A reservation-based traffic management system is implemented at corridor intersections and elevator lobbies, where robots apply for and reserve time slots to prevent conflicts.
(5) Queue Management: The charging station queuing system employs a first-come-first-served (FCFS) strategy with preemptive scheduling for high-priority tasks. The system can be modeled as an M/M/c queuing model, characterized by: server count (robots): $c = 10$; average task arrival rate during peak hours: $\lambda = 42$ tasks/hour; average service rate per robot: $\mu = 5$ tasks/robot/hour; system utilization rate: $\rho = \lambda / (c\,\mu) = 0.84$.

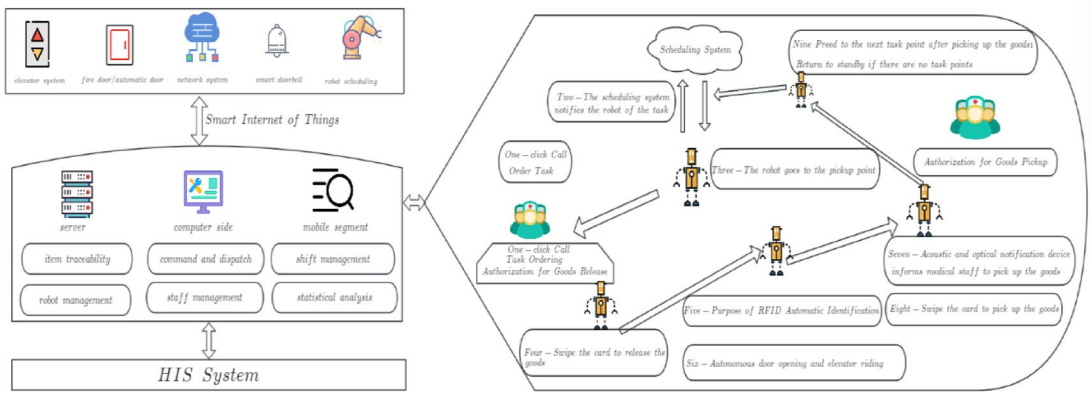


Fig. 2. Pharmacy delivery process.

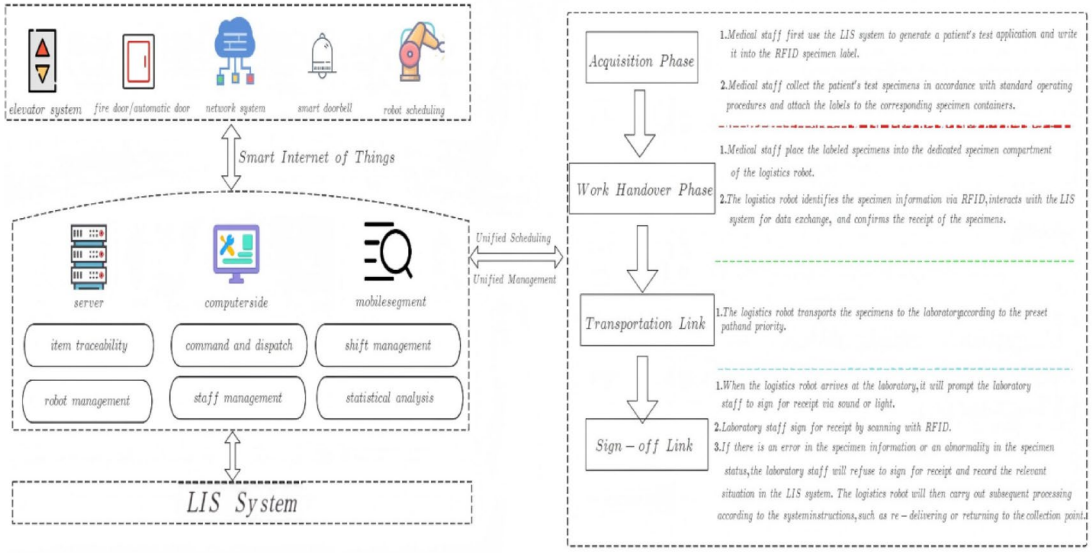


Fig. 3. Specimen delivery process.

Operational modes and workflow

The inpatient pharmacy and sample distribution process incorporates two modes: (1) Scheduled Distribution: Automatically executes timed transportation tasks based on predefined schedules to achieve closed-loop management of routine medication replenishment. (2) On-Demand Distribution: Supports instant delivery triggered by single-point calls via PAD/control consoles to meet emergency medication needs. Both modes can operate concurrently or dynamically switch between them, with no conflicts arising from individual robot task scheduling (See Figs. 2 and 3).

System configuration and technical parameters

The intelligent logistics system consists of 10 autonomous mobile robots (model R0033-A1, SLAM-based AMR) serving 22 open wards across the hospital. The fleet size is determined through peak-hour demand analysis and target service level requirements to ensure timely completion of scheduled and on-demand delivery tasks. Each robot measures 785 mm × 540 mm × 1430 mm, weighs 145 kg, and has a payload capacity of 200 kg, capable of transporting multiple medication boxes or sample containers per trip. Powered by 24 V 30Ah lithium-ion batteries, each unit can operate continuously for approximately 7 h per charge. The system features 10 smart charging stations (one per robot) strategically located near pharmacies, laboratories, and ward corridors, supporting autonomous charging during equipment idleness or when battery level drops below 20%. The total investment amounts to RMB 3.5 million, including 10 robots (unit price RMB 300,000, total RMB 3 million), 10 charging stations (unit price RMB 30,000, total RMB 300,000), and supporting IoT infrastructure (elevator control modules, access control systems, etc.) valued at RMB 200,000.

Robot daily operation, maintenance, and management
The early application of medical AI technology often comes with challenges like unclear management responsibilities and lagging regulatory systems. At this stage, policies and industry standards are not yet fully developed. Technology providers and application units often focus too much on the technology itself, while neglecting the design of regulatory mechanisms for new human-robot collaborative workflows. To address this situation, our hospital established a full-process supervision system when deploying the drug delivery robots. The core of this system includes: (1) Clarifying the responsibilities of all relevant parties and defining human-robot collaboration boundaries; (2) Establishing a library of common fault types and emergency plans to improve risk response capabilities; (3) Developing standardized equipment inspection and maintenance procedures to ensure stable system operation.

Responsibilities of relevant parties in the robot delivery workflow In the regulatory design of the robot delivery workflow, we defined a collaborative responsibility system for multiple parties. At the operational level, the sending party (pharmacy/laboratory) and the receiving party (clinical wards) are the core users of the system. The former is responsible for accurate task dispatch, while the latter takes on the dual roles of receiving goods, managing the on-site environment, and monitoring status. At the management level, a dual-track management mechanism for system and equipment was established. System administrators focus on backend supervision, problem coordination, and technical iteration, while equipment administrators perform preventive maintenance according to standardized plans to ensure hardware stability. At the support level, the hospital's security and IT departments provide technical backing for key infrastructure like access control, surveillance, and networks. Additionally, we provided unified training to all relevant personnel to ensure they master standard operating procedures and emergency response plans, thus building a closed-loop management system with clear responsibilities and rapid response.

Summary of common robot fault types and formulation of emergency plans By recording and summarizing abnormal situations during robot operation, the logistics department identified six common fault types: robot stops when there is an obstacle, robot stops without an obstacle, self-check error after startup, network connection failure, abnormal robot charging, and tablet terminal failure. These faults can generally be resolved through simple operations. Therefore, the system management department classified the fault causes, developed corresponding handling procedures with illustrations, and created a concise and clear emergency plan. This plan assists and guides staff from relevant departments to help the robot recover from unexpected stoppages, ensuring the stability and efficiency of the logistics robot workflow.

Development of robot equipment inspection and maintenance plan To handle hardware wear from daily high-intensity operation, we implemented a tiered preventive maintenance strategy based on component usage frequency. This strategy includes daily or weekly checks for high-frequency key components like wheels and cameras, weekly inspections of environmental factors, and monthly or quarterly planned maintenance for internal core components like LiDAR. For energy management, we set a dual-trigger autonomous charging mechanism at 19:00 daily and when the battery level drops to 20%. Furthermore, the layout of charging piles follows the principle of being close to the starting point, which effectively shortens the robot's empty travel distance, improving operational efficiency and user experience while also reducing the renovation costs of related infrastructure.

Full-time equivalence and data collection

The fleet of 10 robots replaced the original 19 full-time logistics personnel, with a labor efficiency ratio of approximately 1.9 employees per robot. The efficiency improvement stems from the robots' continuous uninterrupted operation capability (without rest, sick leave, or shift rotations) and optimized path algorithms (reducing idle trips and enhancing throughput).

Regarding data collection, trained logistics personnel in the manual group used standardized handheld devices to record task initiation and completion timestamps, with regular audits conducted. The robotic group utilized the dispatching platform for automatic timestamp recording (with second-level precision). A total of 5,792 manual deliveries and 42,430 robotic deliveries were recorded along the "pharmacy → ward" route, while 4,964 manual deliveries and 36,360 robotic deliveries were documented along the "ward → laboratory" route. The higher number of robotic observation instances reflects their superior throughput capacity (See Fig. 4), with all robotic tasks automatically logged by the system.

Statistical analysis

All statistical analyses were performed using SPSS 26.0 and R 4.2.1. Before hypothesis testing, we verified the assumptions of parametric tests. For continuous variables, normality was assessed using Shapiro-Wilk test combined with Q-Q plot visual inspection. Homogeneity of variance between groups was evaluated using Levene's test. For continuous variables meeting both normality and homogeneity assumptions, independent samples t-test was used. Results are presented as mean plus or minus standard deviation with t-value, degrees of freedom, and 95% confidence interval for mean difference. For variables violating these assumptions, non-parametric Mann-Whitney U test was used as an alternative. For categorical or proportional variables, chi-square test was used when expected cell frequency was 5 or greater; otherwise, Fisher's exact test was used. Results are presented as frequency and percentage. To assess the magnitude of observed differences, effect sizes were calculated: Cohen's d for continuous variables and Cramer's V for categorical variables. All primary outcome measures reported 95% confidence intervals. All statistical tests were two-tailed with significance level of 0.05. P-values less than 0.05 were considered statistically significant.

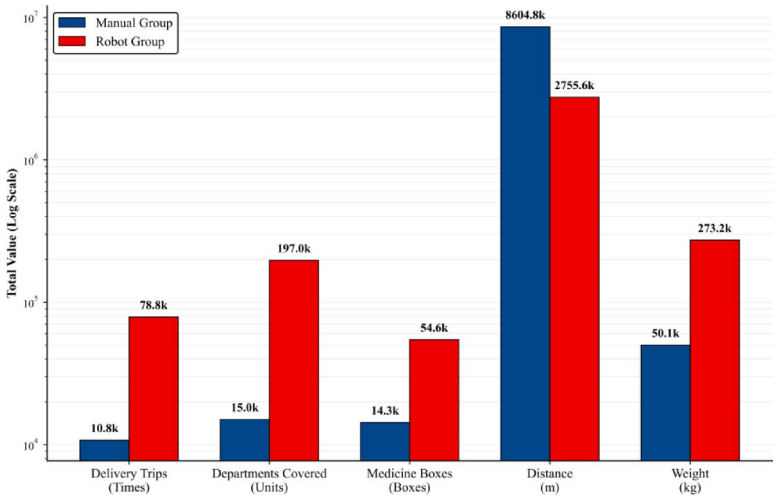


Fig. 4. The two-group workload comparison.

Ethics statement

This study was conducted in accordance with the relevant guidelines and regulations of the Declaration of Helsinki. All methods were carried out in accordance with the approved study protocol and institutional guidelines. The study protocol, including the use of operational logistics data and the involvement of human participants (hospital logistics staff, nurses, and patients), was approved by the Ethics Committee of Shaanxi Provincial People's Hospital (Approval No.: 2025R043).Informed consent was obtained from all human participants involved in this study. Specifically, written informed consent was obtained from all manual delivery personnel (with an average of six months of hospital logistics experience), nurses who participated in satisfaction surveys, and patients who provided satisfaction feedback. For patients, consent was obtained after explaining that only anonymized operational data (delivery times, satisfaction scores) would be collected, with no access to their clinical records or personal health information. For the robotic group, as data were automatically logged by the dispatching platform without identifying individual personnel or patients, the ethics committee approved a waiver of informed consent for those specific data streams, provided that all data remained anonymized.The study only collected operational logistics data such as delivery time, routes, and quantities, without including any direct patient identification information. The robotic system did not access or store patients' clinical data or personal health information. Integration with HIS and LIS only transmitted essential logistics data such as drug/sample identifiers and department codes, without involving patient names or medical record numbers. In terms of data privacy protection, the system implemented a layered protection mechanism for medical item data (e.g., name, specifications, quantity) and transportation process data (e.g., origin/destination locations, timestamps, operator identity). Specific measures included: (1) Encryption of transmission processes using TLS 1.3 protocol and AES-256 algorithm; (2) Storage data was encrypted according to the national SM4 standard, with biometric information independently encrypted after hashing; (3) RBAC model-based implementation adhering to the least privilege principle to ensure only authorized personnel could access necessary data; (4) Complete audit logs were generated for critical operations to facilitate traceability. Additionally, system integration utilized standardized API interfaces and HL7 protocol to transmit only task-required information, reducing privacy leakage risks at the source. All data management processes strictly complied with national standards for medical big data, security, and service management to ensure full-process compliance with patient privacy and sensitive hospital information requirements.

Results

Delivery efficiency analysis (average delivery time)

The robot group was significantly faster than the manual group on both key routes, with time reduced by approximately 32% to 36% ($p < 0.001$) (See Table 2).

Monthly performance trend analysis

To evaluate system stability and potential learning effects, we analyzed monthly delivery time trends for the robot group. Data showed a brief adaptation period in the first month (average delivery time about 14% higher than subsequent months), after which performance stabilized. This indicates the system can achieve rapid maturity within 4–6 weeks after deployment (See Table 3).

Delivery time distribution analysis

The robot group not only had significantly lower average delivery time but also better performance consistency. For the pharmacy-to-ward route, the robot group had a standard deviation of 4.2 min and interquartile range of 4 min, significantly smaller than the manual group's 6.2 min and 8 min. The robot group completed 95% of delivery tasks within 18–28 min, while the manual group's 95% of tasks were distributed between 22 and 44 min,

Metric	Manual group	Robot group	95% CI	Cohen's d	*p*-value
Pharmacy to Ward (min)	31 ± 6.2	21 ± 4.2	8.5–11.5	1.89	< 0.001a
Ward to Laboratory (min)	38 ± 7.2	23 ± 4.6	12.8–17.2	2.48	< 0.001a
Omission rate	0.056%	0.004%	–	OR = 14.0	0.011b
Verification accuracy	97%	100%	–	–	< 0.001b
Integrity rate	99%	100%	–	–	< 0.001b
Nurse satisfaction	94%	97%	–	OR = 2.06	0.002b
Patient satisfaction	90%	93%	–	OR = 1.48	0.037b

Table 2. The results of the two-group data comparison. a indicates *t*-test; b indicates Fisher 's precise test.

Month	Delivery trips (times)	Avg time (min)	SD (min)	Error rate (%)	System uptime (%)
June 2025	10,500	23.9	5.8	0.012	94.2
July 2025	12,565	21.0	5.1	0.006	96.8
August 2025	13,112	21.0	4.5	0.004	98.1
September 2025	13,658	21.0	4.3	0.004	98.5
October 2025	14,204	21.0	4.2	0.003	98.7
November 2025	14,751	21.0	4.1	0.003	99.0

Table 3. Monthly performance trend analysis.

with extreme delay events up to 52 min. The route from ward to laboratory exhibited a similar pattern: the robotic group (P5–P95) took 15–31 min, while the manual group required 26–50 min. These results indicate that the robot system not only outperforms manual delivery in average efficiency but also maintains more stable performance under various operating conditions.

Delivery quality comparison (omission rate, verification accuracy, integrity rate)

The robot group achieved 100% in both verification accuracy and integrity rate, significantly better than the manual group ($p < 0.001$). The omission rate was also significantly lower ($p = 0.011$) (See Table 2).

Workload comparison (delivery trips(times), departments covered (units), medicine boxes (boxes), distance (m), weight (kg))

Results showed the robot system significantly outperformed manual delivery in all dimensions ($p < 0.001$). Specifically, the robot group completed 7.3 times more delivery trips than the manual group; covered 13.1 times more departments; transported 3.8 times more medicine boxes; reduced total transport distance by 68%; and transported 5.5 times more total weight. This demonstrates that the robot system has significant advantages in task processing capacity, service coverage, and transport efficiency.

User satisfaction comparison

Both nurse and patient satisfaction scores were significantly higher for the robot group than the manual group ($p < 0.05$). This indicates that the robot service model gained wider acceptance (See Table 2).

Long-term economic benefits comparison

Figure 5 presents a 10-year cost-benefit comparison analysis between logistics robots and manual delivery models. Based on operational data from our hospital's 500-bed capacity and 22 open wards, manual delivery requires 19 staff members with total annual costs of 1.102 million yuan (including basic salary of 45,000 yuan/person/year, team management costs of 8,000 yuan/person/year, and operational management costs of 5,000 yuan/person/year). At an average annual growth rate of 3%, cumulative labor costs over 10 years reach 12.633 million yuan. The initial investment for the robotic system is 3.5 million yuan, with maintenance rates gradually increasing from 5% in the second year to 8% starting from the fifth year, resulting in total equipment costs of 5.705 million yuan over 10 years. Analysis shows that the robotic system can achieve cumulative cost savings of 6.928 million yuan over 10 years, with an average annual savings of 693,000 yuan and a payback period of 3.4 years.

Sensitivity analysis of cost-benefit model

To evaluate the robustness of economic predictions, we conducted sensitivity analysis on key parameters. In the conservative scenario, 10-year net savings reached 6.31 million RMB. In the baseline scenario, savings reached 6.928 million RMB. In the optimistic scenario, savings reached 7.28 million RMB. The robot system-maintained cost-effectiveness advantages in all three scenarios, indicating investment decisions have strong resilience to parameter fluctuations (See Table 4).

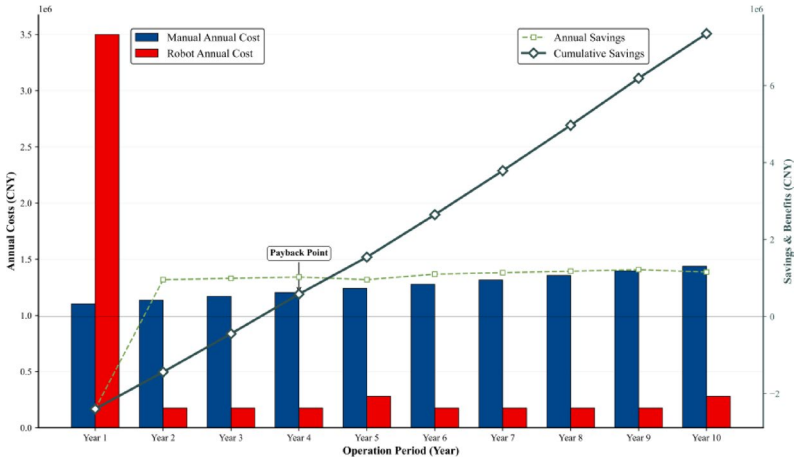


Fig. 5. Cost savings analysis.

Parameter scenario	Conservative	Baseline	Optimistic
Labor cost growth rate	2%/year	3%/year	4%/year
10-year labor cost (Million CNY)	11.85	12.63	13.52
10-year robot cost (Million CNY)	6.49	5.705	4.91
Net savings (Million CNY)	**5.36**	**6.928**	**8.61**
Maintenance cost escalation			
Cost multiplier	+ 20%	0% (baseline)	− 1%*
10-year robot cost (Million CNY)	6.846	5.705	5.648
Net savings (Million CNY)	**5.784**	**6.928**	**6.982**
System downtime impact			
Annual downtime	5%/year	3%/year	2%/year
Net savings (Million CNY)	**6.31**	**6.928**	**7.28**

Table 4. Sensitivity analysis of cost-benefit model. Note:− 1%* represents assumed efficiency gains over baseline; Significance value bold.

Discussion

This study systematically compared two intra-hospital material delivery models through six months of parallel observation. Results clearly show that the logistics robot system demonstrated significant advantages in delivery efficiency, quality control, task capacity, user satisfaction, and long-term cost-effectiveness compared to traditional manual delivery.

Delivery efficiency advantages

The logistics robot distribution model adopted by our hospital demonstrated superior performance during stable operation. Data showed that the average delivery time of the robotic group was reduced by 32% to 36% compared to the manual group ($p < 0.001$). This difference highlights the technical advantages of automated path planning and uninterrupted operation. Manual delivery is inevitably affected by physiological fatigue, subjective path selection, and interruptions such as conversations or elevator waits. In contrast, robots can strictly adhere to optimal path algorithms, enabling standardized 24/7 continuous operation[33,34]. This efficiency improvement particularly reduces waiting times in clinical departments during peak hours or emergency material allocation scenarios, creating valuable time windows for medical work.

Notably, the monthly performance trend analysis revealed learning effects during the initial deployment phase. Data shows delivery times in the first month were approximately 14% higher than subsequent months, after which performance stabilized. This pattern aligns with the typical maturity curve for new technology deployment, indicating rapid system maturation within 4–6 weeks post-deployment. These findings hold significant practical implications for hospital planning robot deployment strategies. It is recommended to reserve an adaptation period during initial implementation and deploy technical support teams to closely monitor system performance, ensuring a smooth transition.

The distribution analysis of delivery times further validates the performance stability of the robotic system. 95% of delivery tasks in the robotic group were completed within 18–28 min, while the manual group's 95% of tasks fell within the 22–44-minute range, with extreme delays observed (maximum 52 min). This variance

reduction, from a six-sigma quality management perspective, demonstrates that the robotic system has achieved process standardization, significantly reduced process variability, and enhanced service consistency.

Delivery quality advantages

In terms of delivery quality, the robotic system achieved 100% verification accuracy and 100% item integrity rates, with an omission rate significantly lower than manual operations ($p = 0.011$). This outcome is primarily attributed to the closed-loop verification mechanism integrated into the robotic system. For instance, scanning QR codes or RFID tags for pickup and receipt confirmation eliminates human errors at the technical level, preventing issues such as misdelivery to incorrect departments, incorrect item retrieval, or missed handovers. This design embodies the core principle of "Poka-Yoke" in lean management: preventing errors through technical measures before they occur, rather than relying on manual inspections for post-event corrections. In contrast, manual operations' accuracy heavily depends on staff accountability, experience, and current working conditions, with relatively higher error risks during personnel rotation and high-intensity work environments[33]. The robotic system transforms delivery processes from reliance on human experience to adherence to technical standards, ensuring precise material flow and traceability—a critical advantage for managing high-value, high-risk items such as pharmaceuticals and specimens.

Algorithm framework and operations research contribution

The algorithm framework of this system integrates classic theories from operations research and robotics, providing a replicable technical paradigm for hospital logistics automation. Path planning uses a hybrid approach: global path based on Dijkstra's algorithm for shortest feasible path, local navigation using A* algorithm with dynamic obstacle avoidance. This layered architecture ensures global optimality while having real-time response capability to environmental changes. Collision avoidance uses DWA, recalculating velocity commands at 10 Hz frequency to ensure safe operation in complex crowd environments. From an operations research perspective, this system models hospital logistics as a Capacitated Vehicle Routing Problem with Time Windows (CVRPTW). The task priority mechanism (urgent > routine > scheduled) corresponds to priority constraints in VRP literature, ensuring emergency supplies are delivered first. Queuing theory analysis revealed key performance indicators of system operation. Based on the M/M/c model ($c = 10$ robots), peak hour system utilization of 0.84 is in a reasonable operating range (usually recommending less than 0.9 to avoid excessive congestion). This model provides theoretical basis for fleet size planning: if demand grows to 50 tasks per hour, maintaining utilization less than 0.9 would require increasing to 12 robots.

Task capacity and scalability analysis

During the six-month observation period, the robot system completed significantly more delivery tasks, covered more departments, and transported more materials than the 19-person manual group. This fully demonstrates the powerful scalability and high-throughput processing capability of the robot system. More importantly, this result shows that logistics robots can free up valuable human resources from low-value repetitive physical labor. Currently, healthcare systems worldwide face challenges of insufficient nurse staffing, heavy workloads, and job burnout[5,35,36]. A large amount of nurse time is consumed on non-core nursing tasks[37,38]. Introducing a logistics robot system is equivalent to equipping the clinical frontline with a diligent, precise, and efficient "logistics support force," allowing nurses to focus on professional work such as patient assessment, treatment implementation, and health education. From a scalability perspective, SLAM-based navigation systems require only basic infrastructure modifications, demonstrating significant advantages over fixed rail systems that demand large-scale infrastructure investments. The study's configuration ratio of 10 robots serving 22 wards (approximately 1:2.2) provides a reference for fleet sizing planning in other hospitals. The scheduling platform supports dynamic task allocation, with scalability tests showing the system can efficiently handle 15–20 concurrent tasks, validating technical feasibility for deployment in large-scale hospitals.

User satisfaction analysis

Notably, both nurse and patient satisfaction were significantly higher in the robot group ($p < 0.05$). This result breaks the traditional stereotype that automated services lack human care. Analysis shows that the timeliness, accuracy, and predictability of robot services significantly improved user experience[39,40]. For nurses, reliable robot service can reduce anxiety and extra communication costs caused by material delays or errors. For patients, faster service response also indirectly improves their healthcare experience. This result indicates that automated service processes can gain wide user acceptance and trust.

Long-term economic benefits and sensitivity analysis

The economic model analysis of this study not only demonstrates the operational efficiency and quality advantages of the logistics robot system but also reveals its huge long-term financial value. Although the initial purchase cost of the robot system was as high as 3.5 million RMB, from a 10-year lifecycle perspective, its total cost of about 5.705 million RMB is far lower than the continuously growing labor cost of 12.633 million RMB, with cumulative savings exceeding 6.928 million RMB. Sensitivity analysis further verified the robustness of economic predictions. In all three scenarios, the robot system-maintained cost-effectiveness advantages, indicating that investment decisions have strong resilience to parameter fluctuations. This finding has important decision-making reference value for hospital managers, even in environments with high economic uncertainty, robot investment remains a financially sound choice. Labor costs exhibit inherent rigidity and will continue to rise in tandem with increasing average social wage levels. In contrast, the primary cost of robotic systems lies in initial investment, while subsequent operational and maintenance expenses remain relatively manageable. This upfront investment coupled with long-term returns aligns with the fundamental principles of large-scale fixed

asset investments. When evaluating such projects, it is essential to go beyond initial purchase prices and adopt a holistic approach through Total Cost of Ownership (TCO) analysis or cost-benefit assessment frameworks.

External validity and generalizability

From the perspective of external validity, SLAM-based navigation systems require an initial mapping phase (typically taking 1–2 weeks for medium-sized hospitals) and only basic infrastructure modifications. Compared to fixed track systems that demand large-scale infrastructure investments, this solution demonstrates significant advantages. The logistics robots in this study possess autonomous navigation capabilities and adaptability to various floor layouts, including campus environments composed of multiple buildings. However, system performance may vary depending on hospital infrastructure variations. Insufficient corridor widths (< 1.5 m) may limit bidirectional passage capacity; limited elevator capacity or congestion during peak hours could become bottlenecks; and legacy hospital information systems may require customized interface development. Potential deployment hospitals are advised to conduct infrastructure assessments during planning stages, including corridor width measurements, elevator capacity analysis, and information system compatibility testing. Additionally, the system employs standardized HL7 interfaces and API protocols, enabling seamless integration with most commercial hospital information systems. This open architecture design reduces integration barriers and facilitates replication across different healthcare institutions.

Limitations

This study has several limitations. First, as a single-center study, results may be influenced by our hospital's specific building layout, department distribution, management processes, and personnel culture. Caution is needed when generalizing to other hospitals. Second, although we controlled for key variables such as routes and time periods, confounding factors such as real-time elevator congestion and instantaneous workload fluctuations were not systematically recorded, which may affect results to some extent. Additionally, this study did not use Discrete Event Simulation technology to model system behavior under different operating scenarios. Variables such as peak hour elevator wait times and charging station queue dynamics were evaluated through actual post-deployment data rather than pre-deployment simulation analysis. The algorithm framework is based on several idealized assumptions, such as task arrivals following Poisson distribution and service times following exponential distribution. In actual operation, task demand may present more complex patterns, and the algorithm adaptability in these situations needs further verification. Future research can proceed in the following directions: (1) multi-center validation: conduct comparative studies across medical institutions of different sizes, layouts, and management processes to establish more universal performance benchmarks. (2) Confounding factor control: use more refined data collection schemes to record dynamic variables and achieve more precise effect estimation through statistical methods. (3) Pre-deployment simulation optimization: introduce discrete event simulation technology to model peak hour bottlenecks before hardware investment. (4) AI-driven dynamic scheduling: integrate machine learning algorithms for demand prediction and real-time task optimization. (5) Human-robot collaboration optimization: explore optimal workflow design that maximizes the complementary advantages of human flexibility and robot precision. (6) Application scenario expansion: extend the robot system to sterile material transport, medical waste management, meal delivery, and other logistics areas.

Implications for hospital management

Intelligent logistics systems should be viewed as strategic investments to enhance hospital core competitiveness, not merely as cost centers. Their value is reflected in multiple dimensions including operational efficiency, medical quality, employee satisfaction, and long-term economic benefits. Implementing a robot system is not simply about replacing manpower, but a process of redesigning workflows to achieve human-robot collaborative optimization. In terms of economic decision-making, sensitivity analysis shows that robot investment maintains cost-effectiveness advantages under different economic scenarios, providing decision confidence for hospital managers. We recommend adopting a Total Cost of Ownership analysis perspective when making investment decisions, rather than focusing only on initial investment amount. In terms of infrastructure planning, we recommend hospitals conduct systematic assessment before deployment, including corridor width measurement (recommended 1.5 m or more to support bidirectional traffic), elevator traffic capacity analysis, and information system compatibility testing. In terms of personnel management, hospitals need to develop supporting personnel transition training plans to help employees adapt to the new model of collaborating with robots. Transport personnel can transition to positions such as equipment maintenance, system scheduling, or emergency response, realizing the appreciation of human capital value. For large hospitals, a phased regional promotion strategy can be adopted to introduce robot systems, first verifying effects in pilot departments before full promotion. In operational management, it is essential to establish a continuous data monitoring and evaluation mechanism to dynamically optimize system configurations and scheduling strategies based on actual operational outcomes. Implementing Key Performance Indicator (KPI) dashboards is recommended to track core metrics such as delivery timeliness, system utilization rates, and failure rates in real-time, thereby providing data-driven support for management decisions.

Conclusions

Through a six-month empirical comparison of manual and robot-assisted intra-hospital material delivery models, this study systematically confirmed that the logistics robot system has significant comprehensive advantages in delivery efficiency, quality, scalability, user satisfaction, and long-term cost-effectiveness. The system achieved closed-loop multi-task management of intra-hospital logistics. A fleet of 10 robots can complete about 440 delivery tasks daily in 22 wards, equivalent to replacing 19 full-time logistics personnel with a labor

efficiency ratio of 1.9 workers per robot. From a methodological perspective, this study combined operations research theory with robotics technology, providing a replicable technical paradigm for hospital logistics system optimization. The system utilization analysis based on M/M/c queuing model provides theoretical basis for fleet size planning. Sensitivity analysis confirmed the robustness of economic predictions, maintaining cost-effectiveness advantages in all three scenarios. The research results provide solid evidence-based basis for intelligent logistics transformation in similar medical institutions, indicating that introducing automated logistics technology is a key measure for achieving refined hospital management, optimizing resource allocation, and improving overall service levels.

Data availability

The data generated and/or analyzed during the current study are not publicly available due to restrictions based on patient privacy and hospital operational confidentiality. However, anonymized data that support the findings of this study are available from the corresponding author Xiaomei Liu upon reasonable request.

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Author contributions

Conceptualization, Methodology, Supervision, Project Administration, Funding Acquisition, writing—Review & Editing: X.L., and Y.S.. Investigation, Data Curation, Formal Analysis, Writing—Original Draft: M.L. and L.L. Software, Validation: Y.G. and P.L. Resources, Investigation: L.Z. and M.W.. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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